

A Local Hybrid Pipeline for Nuclei Microscopy Image Analysis

VLM Description, Classical Features, and U-Net Segmentation

[GitHub](#)

1. Introduction and Setup

This report documents the development of a hybrid biomedical image-analysis pipeline built for a fluorescence microscopy nuclei-segmentation dataset (80 train / 20 validation / 12 test images, 256×256 RGB, blue-channel DAPI-style nuclear stain, with matching binary masks). The pipeline processes each image sequentially through four stages: raw image → segmentation → quantitative region features → structured JSON record → short narrative, all of which are performed with vision-language model (VLM) via Ollama, classical image processing (scikit-image), and a U-Net trained from scratch in PyTorch. The Pipeline uses the LLaVA model via Ollama.

2. Task 1: Data Preparation and VLM Description

Images were converted to grayscale and resized to 256x256 (no-op here as the source images were already at that size, but done explicitly for pipeline robustness). A set of sample images and an intensity histogram were generated for EDA (in the accompanying notebook).

Naive prompt: “What do you see in this image?”

Naive output: the model misread the image entirely, describing it as “a night sky background with a pattern of circular or spherical shapes... stars, planets... most likely artificial... a digital display.” This demonstrates that an unconstrained prompt gives the VLM no grounding in the imaging domain, producing a confidently wrong analogy rather than an “uncertain” response.

Optimized prompt (used verbatim, recorded in full as required):

“You are an image description assistant helping catalogue microscopy images for research purposes. You are NOT a diagnostic tool and must not provide any clinical or diagnostic interpretation. Describe ONLY what is visually observable in this fluorescence microscopy image. Do not guess at disease state, patient information, or clinical significance. If you are not confident about any field, respond with ‘uncertain’ for that field rather than guessing. Respond ONLY with a valid JSON object in exactly this format, with no extra text before or after it: { “modality”: ..., “tissue_type”: ..., “notable_features”: ..., “image_quality”: ... }”

Optimized output (run 1): {"modality": "fluorescence microscopy", "tissue_type": "cells", "notable_features": "small, spherical, fluorescent objects... uniformly sized and shaped... dark background...", "image_quality": "good"}

Optimized output (run 2, identical input): {"modality": "Fluorescence microscopy", "tissue_type": "Cell culture", "notable_features": "...blue-stained cells... staining appears uneven... noticeable noise...", "image_quality": "Adequate"}

The two runs are not identical (confirmed programmatically, Identical outputs = False), differing in tissue_type phrasing and the quality assessment (“good” vs “adequate”), demonstrating expected non-determinism in local LLM sampling even with a structured, constrained prompt. The optimized prompt correctly anchored the model to the domain (fluorescence microscopy, cells) where the naive prompt failed completely.

3. Task 2: Classical Features and Numbers-First Interpretation

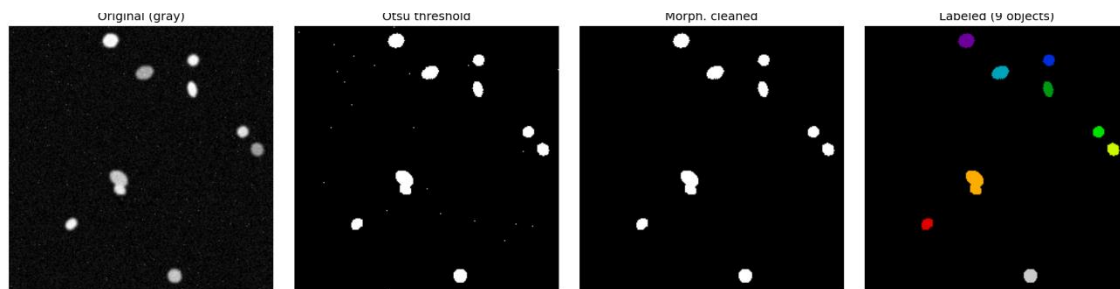


Figure 1: Classical segmentation pipeline: grayscale → Otsu threshold → morphological cleanup → labeled components (9 objects).

The representative image had 9 objects detected by Otsu thresholding, removal of small objects and binary closing, and connected-component labeling (full per-object table in the notebook): mean area 144.6 px, mean eccentricity 0.50, mean solidity 0.94, mean intensity 0.26. This numeric summary (without image) was passed to the LLM, which returned: {"n_objects": 9, "density_class": "moderate", "shape_regularity": "regular", "quality_flag": "none"}.

Comparison with Task 1: the numbers-first description was precise and reproducible when applied to countable geometry (exact number of objects, regularity from eccentricity/solidity) and lacked awareness of information not embedded in the summary statistics: that there was visible speckle/noise in the raw image that was not flagged as a quality concern, and that object #7 in the labeled image was really two touching nuclei combined into a single connected component (visible directly in the image but invisible in an aggregate eccentricity value). The direct VLM description (Task 1) was also able to report qualitative visual context

(staining pattern, noise) not reported by the numbers-first path, but was inconsistent across runs and provided no quantitative counts. This trade-off is discussed further in Question 1 (Section 7).

4. Task 3: U-Net Segmentation

A U-Net was implemented from scratch in PyTorch (encoder/decoder with skip connections, feature widths 32–64–128–256, double-conv blocks with batch norm). Loss combined BCE and Dice loss; Adam optimizer, lr=1e-3, batch size 4, 30 epochs.

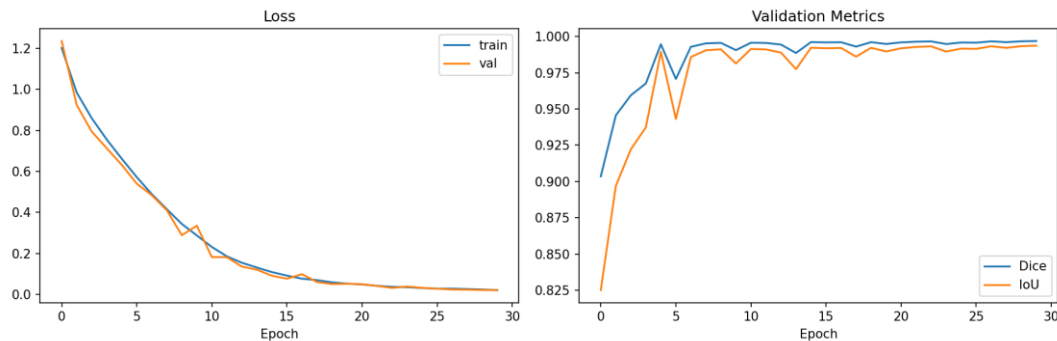


Figure 2: Training/validation loss (left) and validation Dice/IoU (right) across 30 epochs.

Final epoch (30/30): train_loss=0.0212, val_loss=0.0210, validation Dice=0.9968, validation IoU=0.9937. Train and validation loss track closely throughout, with no evidence of overfitting on this small dataset.

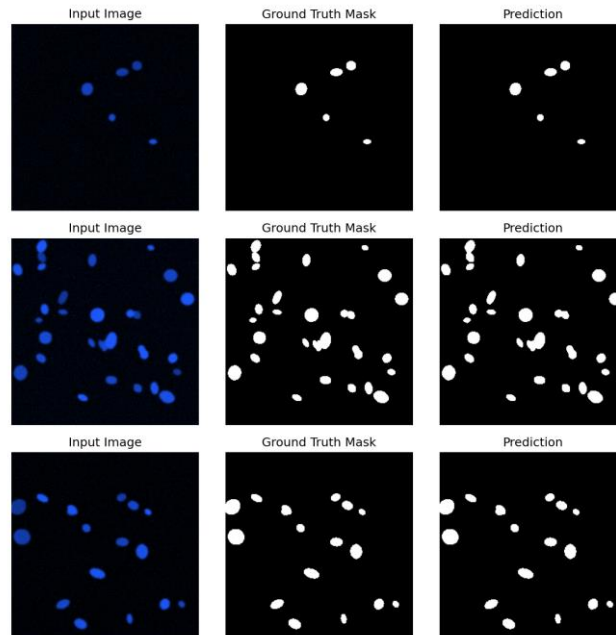


Figure 3: Input / ground-truth mask / U-Net prediction for three validation images, including a visually denser example in the middle row.

The model reproduced ground-truth masks closely across three examples, including the visually denser middle example where most individual nuclei were separated successfully and small dim nuclei were recovered. Dice/IoU this high (>0.99) should be interpreted cautiously: with only 20 validation images of high-contrast, well-separated nuclei on a clean dark background, the task is considerably easier than typical clinical microscopy with variable staining density, dense clustering, and higher noise the metric reflects strong fit to this specific, relatively clean dataset rather than proof of general clinical readiness.

5. Task 4: Hybrid Pipeline on Test Set

For each of the 12 unseen test images: U-Net mask → regionprops feature table → llama JSON record (image_id, n_objects, mean_area, density_class, quality_flag) → one-paragraph narrative. All records were aggregated into a pandas DataFrame and exported as task4_results.csv.

image_id	n_objects	mean_area	density_class	quality_flag
test_000.png	8	191.2	dense	none
test_001.png	14	200.5	dense	none
test_002.png	26	219.7	moderate	none

test_003.png	19	203.9	dense	none
test_004.png	43	328.7	dense	none
test_005.png	16	406.9	dense	none
test_006.png	8	198	moderate	none
test_007.png	23	205.7	moderate	none
test_008.png	23	239.8	moderate	none
test_009.png	21	178.9	dense	none
test_010.png	39	335.1	moderate	none
test_011.png	20	184	dense	none

Table 1: Task 4 aggregated hybrid-pipeline results across the 12 test images

Example narrative (test_004.png, 43 objects, mean area 328.7 px): “The image is a microscopy image of a segmented sample containing 43 detected objects with an average area of approximately 329 pixels, consistent with a dense field of nuclei.”

A data-quality issue was found and corrected during development: llama initially echoed the placeholder syntax from the prompt template verbatim (e.g. density_class = “<sparse | moderate | dense, or uncertain>”) instead of selecting one option. This was fixed by (a) rewriting the prompt to give an explicit enumerated instruction plus a worked example instead of angle-bracket placeholders, and (b) adding a rule-based fallback in code (classifying density from n_objects thresholds) that only activates if the LLM output is not one of the four valid categories. After the fix, all 12 records parsed to valid categories. A residual reliability issue remains, however: density_class is not perfectly monotonic in n_objects across images (e.g. test_000 with 8 objects was labelled “dense” while test_002 with 26 objects was labelled “moderate”), showing that even numerically-grounded LLM judgments on ordinal categories should be treated as advisory rather than authoritative (see Question 4).

6. Extension: Robustness Under Image Corruption

Two test images (test_000, 8 detected objects; test_004, 43 detected objects) were each subjected to two corruption types: heavy blur and low contrast and passed through the pipeline to identify the earliest stage at which each corruption becomes detectable.

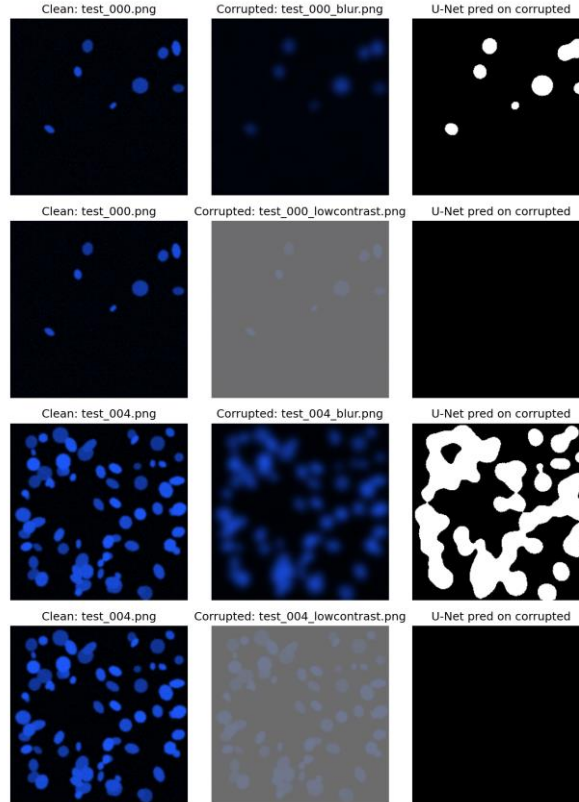


Figure 4: Clean input vs corrupted input and U-Net predicted mask for the corrupted image under blur and low-contrast perturbations of two source test images.

Source	Corruption	Clean n_obj	Corrupt n_obj	Clean area	Corrupt area
test_000.png	blur	8	7	191.2	426.0
test_000.png	low contrast	8	0	191.2	0.0
test_004.png	blur	43	10	328.7	2791.4
test_004.png	low contrast	43	0	328.7	0.0

Table 2: Object count and mean area, clean vs corrupted, by corruption type.

Two distinct failure modes were identified. Under blur, the U-Net does not fail silently: it merges adjacent nuclei into fewer, larger connected regions (test_004: 43→10 objects; mean area increased roughly 8.5×, 328.7→2791.4), producing plausible-looking but structurally incorrect masks the earliest reliable detection point is the regionprops feature table (an implausible area jump), not the mask image itself, which still looks like clean blob shapes. Under low contrast, the U-Net fails completely (0 detected objects in

both cases): its learned intensity threshold never fires below the contrast range seen in training, so this corruption is detectable immediately at the mask stage. The downstream quality mechanism did not identify low contrast as a meaningful segmentation failure. Although one low contrast case triggered the generic invalid LLM output fallback flag, no explicit quality flag indicating an empty or failed segmentation was produced.

7. Reflection Questions

1. Which description is more useful/trustworthy: direct VLM (Task 1) or numbers-first (Task 2)?

The numbers-first description is more trustworthy for anything quantitative: object counts and shape statistics are deterministic given a fixed segmentation, reproducible, and directly traceable to pixel data. The direct VLM description is more useful for qualitative visual context (staining pattern, background noise, overall gestalt) that never gets encoded into the numeric summary, but it is less trustworthy it varied between identical runs and, in Task 1, initially misread the image with no numeric grounding at all. In practice the two are complementary: numbers-first for anything that will be counted, measured, or logged; VLM description only as an auxiliary qualitative note, never as the sole source of a quantitative claim.

2. Did the U-Net improve on classical Otsu segmentation? Give one example each way.

Yes, clearly, on the densely-populated images. In the representative Task 2 image, Otsu + morphological cleanup merged two touching nuclei into a single connected component (object #7 in Figure 1, area 286 px roughly double the 144.6 px mean), an error the U-Net's learned features avoided in the visually denser validation example in Figure 3 (middle row), where most individual touching nuclei were separated correctly. Otsu can still do as well as, or better than, the U-Net on very sparse, high-contrast, well-separated images (e.g. the 8-object cases), where a simple global threshold is already close to optimal and there is little for a learned model to improve on.

3. Report Dice and IoU. What do they mean, and where does the model make mistakes?

Final validation Dice = 0.9968, IoU = 0.9937. Dice measures the overlap between predicted and ground-truth masks ($2 \times \text{intersection} / \text{sum of areas}$); IoU is the stricter intersection-over-union of the same two regions both close to 1.0 indicate near-complete agreement with ground truth on this validation split. The clearest mistakes are in dense, tightly-clustered regions, where nearby nuclei are prone to merging into a single predicted blob rather than being separated (visible faintly even in the well-performing middle example of Figure 3); the robustness test in Section 6 shows this same merging failure mode becomes dramatically worse under blur corruption.

4. Where can the LLM hallucinate, and what design choices reduce that risk?

Hallucination risk is present at each LLM call: the naive Task 1 prompt hallucinated a “night sky” interpretation with no grounding; the optimized Task 1/2/4 prompts had non-deterministic wording and, in one, literally repeated the form of the prompt placeholders as if they were a value. The following are design decisions taken to mitigate this risk: (a) force the structured JSON output; (b) explicitly allow “uncertain”, giving the model a safe non-answer; (c) use the deterministic regionprops measurements for numeric fields (n_objects, mean_area, JSON quality checks) instead of relying on the LLM's own counting; (d) a rule-based fallback to handle invalid LLM categorical output. The structured JSON is the “source of truth” because it is only this part of the record that is directly traceable back to the pixel measurement; the narrative portion is free-form language generated from the structured JSON, and should never be taken as an independent source of truth, or as a more reliable source if the structured JSON is incorrect, the narrative will simply provide a fluent description of the incorrect information..

5. Would you trust any part of this system clinically? What single change would most improve trustworthiness?

None of this system as constructed should be relied upon in a truly clinical setting. The dataset is small (80/20/12 images), synthetic in appearance, and exceptionally clean, compared to real clinical microscopy, for this purpose; the VLM used is not part of the specified model for infrastructure reasons; and the robustness test reveals that the pipeline fails silently under two very common image quality problems (blur, low contrast) without any downstream flag being set. The one thing that would strongly contribute to the trustworthiness would be to add an automatic sanity check that would check the predicted object count/mask area against expected ranges (or a classical Otsu cross-check) and would trigger the quality_flag to a hard failure state (not “none”) where the two disagree significantly or the mask is empty and would bring the known silent failure mode from Section 6 into the public view before any downstream narrative.

References

- Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI 2015.
- Otsu, N. (1979). A Threshold Selection Method from Gray-Level Histograms. IEEE Transactions on Systems, Man, and Cybernetics, 9(1), 62–66.
- van der Walt, S. et al. (2014). scikit-image: image processing in Python. PeerJ, 2:e453.
- Ollama (2024/2026). Local LLM/VLM runtime. <https://ollama.com>